WeatherWise Analytics

A MongoDB-Powered Weather Intelligence and Machine Learning System

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1. Abstract & Executive Summary

WeatherWise Analytics presents a complete machine learning and database intelligence platform engineered to store, query, and analyze 73+ years of daily Delhi weather observations (1951 to 2024, 26,803 records). Coupling a persistent MongoDB NoSQL database with Scikit-learn pipelines, the platform evaluates Logistic Regression, Decision Tree, and Random Forest models for next-day precipitation prediction using leak-free chronological evaluation. An integrated Random Forest regressor predicts next-day maximum temperature (R^2 = 0.944, MAE = 1.18°C). A rules-based decision engine translates statistical probabilities into actionable recommendations.

2. Dataset & Data Audit

Metric	Value
Total Daily Records	26,803
Temporal Range	1951-01-01 to 2024-06-19
Observation Span	73.5 Continuous Years
Non-Rainy Days (Class 0)	22,093 (82.6%)
Rainy Days (Class 1)	4,658 (17.4%)
Duplicate Dates	0 (Zero duplicates detected)

3. System Architecture & PyMongo Integration Classification

WeatherWise Analytics leverages the official PyMongo 4.x driver to interface with MongoDB Community Server (database: 'weatherwise_db', collection: 'daily_weather'). PyMongo functionality is classified into seven key roles:

• **Client Pooling & CSOT:** MongoClient initialized with serverSelectionTimeoutMS=3000 and connectTimeoutMS=3000 to prevent application stalling if unhosted.

• **Diagnostic Health Pings:** Verifies connectivity via client.admin.command('ping') with visual UI indicators.

• **Idempotent Bulk Ingestion:** Processes 26,803 records in 1,000-doc chunks using UpdateOne({'Record_Key': key}, {'$set': doc}, upsert=True) with bulk_write(ordered=False), ensuring zero duplicate insertions on repeat imports.

• **Indexing Architecture:** Enforces uniqueness on 'Record_Key' (YYYY-MM-DD), ascending chronological index on 'Date_Parsed', and compound index on (Year, Month).

• **Optimized Query Formulation:** Uses collection.find() with strict projections and limits for fast data exploration.

• **Collection Statistics:** Uses collection.count_documents({}) and list_indexes() for live telemetry.

• **Offline Fallback Architecture:** Catches ServerSelectionTimeoutError and gracefully activates local CSV mode without simulated connection states.

4. Classification Model Performance

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.7290	0.3799	0.8073	0.5167	0.8412
Decision Tree	0.7477	0.4014	0.8271	0.5405	0.8527
Random Forest	0.7688	0.4230	0.7927	0.5516	0.8506

5. Temperature Regression Results

Regressor Model	MAE (°C)	RMSE (°C)	R^2 Score
Random Forest Regressor	1.17	1.62	0.9455
Decision Tree Regressor	1.21	1.67	0.9422
Linear Regression	1.20	1.68	0.9418

6. Weather Recommendations & Practical Utility

Predictions are translated into understandable suggestions: umbrella advisories at 30% and 60% probability thresholds, thermal guidance for clothing across five temperature regimes, and outdoor activity feasibility indices. All recommendations include explicit notifications regarding their educational, statistical basis.

7. Conclusion

WeatherWise Analytics demonstrates the effectiveness of coupling MongoDB document storage with leak-free machine learning pipelines. Random Forest emerged as the top performer across classification and temperature regression, providing a reliable benchmark for historical weather analysis.